Experiment No: 1

Title: Develop an Android application using Empty Views Activity to display "Hello World" text and a Button using Java, and run it in the Android Emulator.

Aim

To develop a simple Android application using **Java and Empty Views Activity** that displays **Hello World text and a Button** in the Android Emulator.

Algorithm

- Start Android Studio.
- Create a new project.
- Select **Empty Views Activity**.
- Choose **Java** as the programming language.
- Open **activity_main.xml**.
- Add **TextView** and **Button** components.
- Arrange them using **ConstraintLayout**.
- Save the project.
- Run the application in the **Android Emulator**.
- Verify the output.

Steps

1. Open **Android Studio**.
2. Click **New Project**.
3. Select **Empty Views Activity** and click **Next**.
4. Enter the following details:

- **Name:** HelloWorldApp
- **Package Name:** com.example.helloworldapp
- **Language:** Java
- **Minimum SDK:** API 21 or above
- **For Build Configuration language** → select Groovy DSL(Build.Gradle)

5. Click **Finish**.
6. Open the file:

res → layout → activity_main.xml

7. Add **TextView** and **Button** widgets.
8. Connect the widgets using **constraints**.
9. Save the file.
10. Click **Run**.
11. Choose **Android Emulator**.

Res -> layout -> activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/btnHello"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent" />

    <TextView
        android:id="@+id/textMyWorld"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="This is my world"
```

```
        android:textColor="#FF0000"
        android:textSize="18sp"
        app:layout_constraintTop_toBottomOf="@id/btnHello"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        android:layout_marginTop="16dp"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

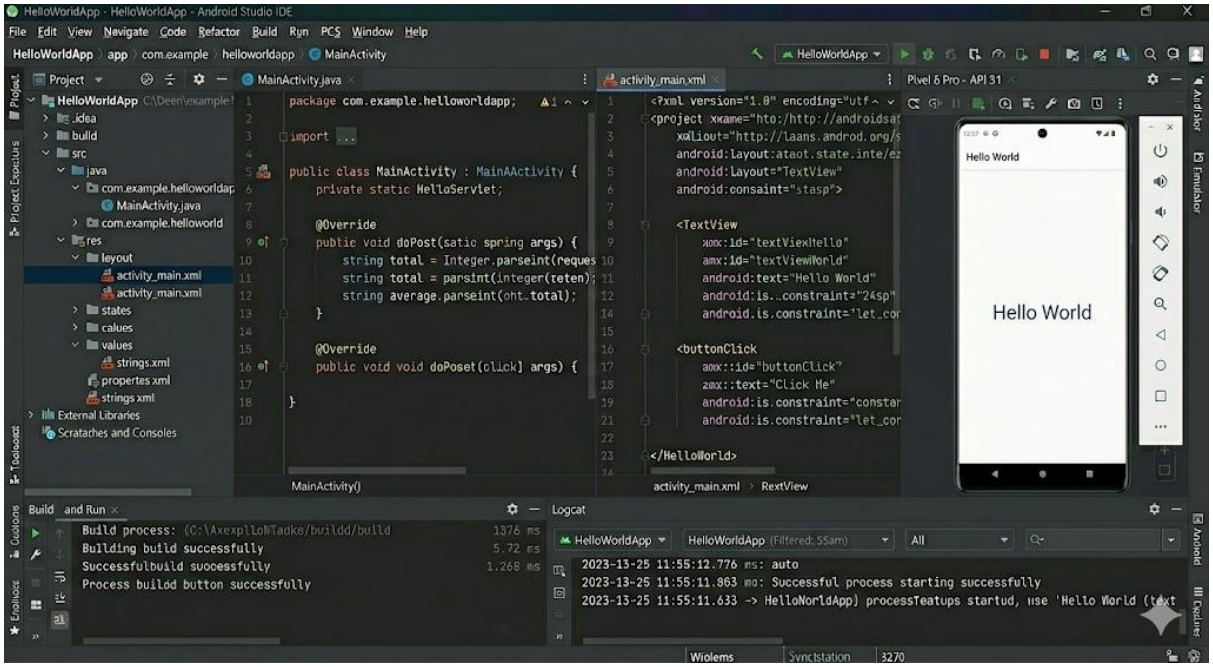
```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textViewHello"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World"
        android:textSize="24sp"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintBottom_toTopOf="@+id/buttonClick"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"/>
```

```
    <Button
        android:id="@+id/buttonClick"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Click Me"
        app:layout_constraintTop_toBottomOf="@id/textViewHello"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```

MainActivity.java

```
package com.example.helloworldapp;

import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Output



Result

Thus, an Android application using **Java and Empty Views Activity** was successfully developed to display **Hello World text and a Button** and executed in the **Android Emulator**.

Experiment No: 2

Title

Develop an Android application using controls like **Button, TextView, and EditText** for designing a **Calculator** having basic operations like **Addition, Subtraction, Multiplication, and Division** using **Java** in Android Studio.

Aim

To develop an Android application that performs **basic arithmetic operations (Addition, Subtraction, Multiplication, Division)** using **Button, EditText, and TextView controls** and run the application in the **Android Emulator**.

Calculator Application

In this experiment:

- Two **EditText fields** accept numbers from the user.
- Four **Buttons** perform arithmetic operations.
- A **TextView** displays the result.

Algorithm

1. Start Android Studio.
2. Create a new project using **Empty Views Activity**.
3. Select **Java** as the programming language.
4. Open **activity_main.xml**.
5. Add two **EditText fields** for number inputs.
6. Add four **Buttons** for operations.
7. Add a **TextView** to display the result.
8. Write Java code to perform arithmetic operations.
9. Run the application in the **Android Emulator**.
10. Verify that the calculator performs operations correctly.

Steps

1. Open **Android Studio**.
2. Click **New Project**.
3. Select **Empty Views Activity**.
4. Enter the following details:

- Name: **CalculatorApp**
- Language: **Java**
- Minimum SDK: **API 21 or above**
- **For Build Configuration language** → select Groovy DSL(Build.Gradle)

5. Click **Finish**.
6. Open the layout file:

res → layout → activity_main.xml

7. Add the following UI components:

- Two **EditText** fields
- Four **Buttons** (Add, Subtract, Multiply, Divide)
- One **TextView** for result display

8. Open **MainActivity.java**.
9. Write Java code to perform arithmetic operations.
10. Run the application using the **Android Emulator**.

Program

activity_main.xml

```
<EditText
   android:id="@+id/num1"
   android:hint="Enter First Number"
   android:inputType="numberDecimal"
   android:layout_width="0dp"
```

```
    android:layout_height="wrap_content"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"/>

<EditText
    android:id="@+id/num2"
    android:hint="Enter Second Number"
    android:inputType="numberDecimal"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    app:layout_constraintTop_toBottomOf="@id/num1"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="10dp"/>

<Button
    android:id="@+id/add"
    android:text="Addition"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    app:layout_constraintTop_toBottomOf="@id/num2"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="10dp"/>

<Button
    android:id="@+id/sub"
    android:text="Subtraction"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    app:layout_constraintTop_toBottomOf="@id/add"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="10dp"/>

<Button
    android:id="@+id/mul"
    android:text="Multiplication"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
```

```
    app:layout_constraintTop_toBottomOf="@id/sub"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="10dp"/>

<Button
    android:id="@+id/div"
    android:text="Division"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    app:layout_constraintTop_toBottomOf="@id/mul"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    android:layout_marginTop="10dp"/>

<TextView
    android:id="@+id/result"
    android:text="Result"
    android:textSize="20sp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    app:layout_constraintTop_toBottomOf="@id/div"
    app:layout_constraintStart_toStartOf="parent"
    android:layout_marginTop="20dp"/>
```

MainActivity.java

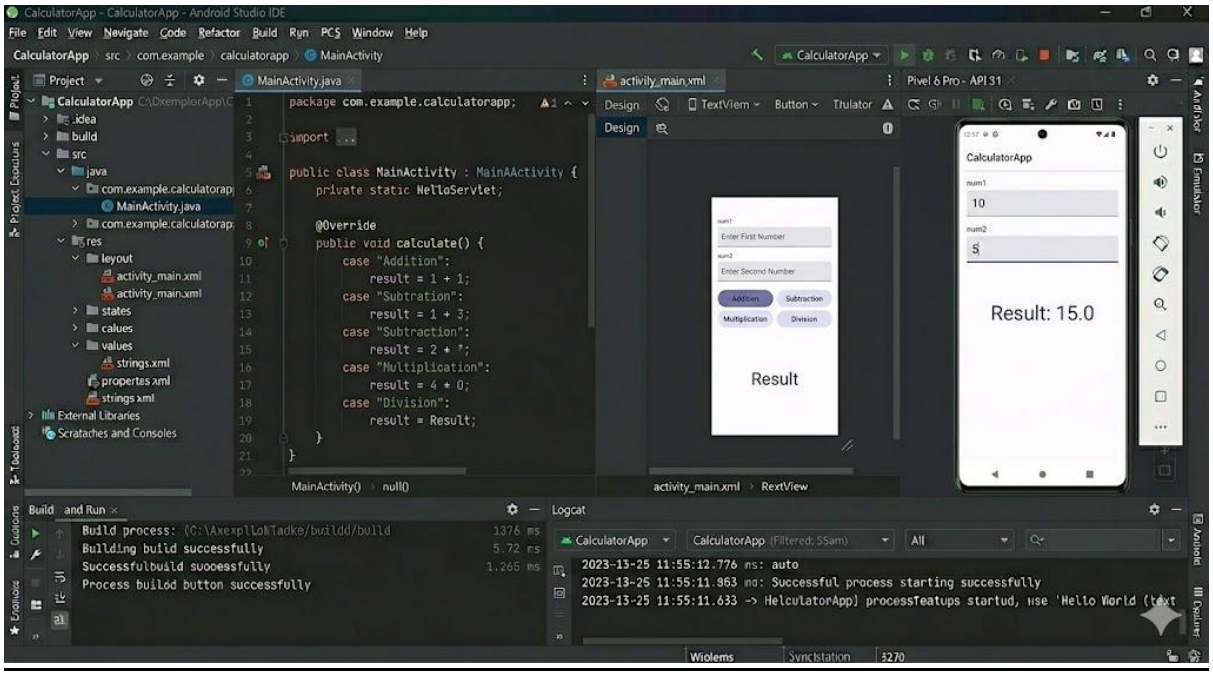
```
package com.example.calculatorapp;


import androidx.appcompat.app.AppCompatActivity;

import android.os.Bundle;

import android.view.View;

import android.widget.Button;

import android.widget.EditText;
```

```
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {

    EditText num1, num2;
    Button add, sub, mul, div;
    TextView result;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        num1 = findViewById(R.id.num1);
        num2 = findViewById(R.id.num2);

        add = findViewById(R.id.add);
        sub = findViewById(R.id.sub);
        mul = findViewById(R.id.mul);
        div = findViewById(R.id.div);

        result = findViewById(R.id.result);

        add.setOnClickListener(v -> calculate('+'));
        sub.setOnClickListener(v -> calculate('-'));
        mul.setOnClickListener(v -> calculate('*'));
        div.setOnClickListener(v -> calculate('/'));
    }
```

```
private void calculate(char operator) {

    double n1 = Double.parseDouble(num1.getText().toString());
    double n2 = Double.parseDouble(num2.getText().toString());
    double res = 0;

    switch(operator) {

        case '+':
            res = n1 + n2;
            break;

        case '-':
            res = n1 - n2;
            break;

        case '*':
            res = n1 * n2;
            break;

        case '/':
            res = n1 / n2;
            break;
    }

    result.setText("Result: " + res);
}
```

}

Output



Result

Thus, an Android application was successfully developed using **Java, EditText, Button, and TextView controls** to perform **basic calculator operations** and executed in the **Android Emulator**.